

Databricks' Delta Lake is now completely open source

Databricks, a cloud data and artificial intelligence (AI) specialist, has announced that it will contribute all enhancements to its Delta Lake platform to the open source community. Databricks was born from open source software, and making more of its features available is seen as a prudent move as the importance of open standards in cloud-based data storage and processing grows.

Delta Lake will be handed over to the open source Linux Foundation, as announced at the Data + AI summit, and all Delta Lake APIs will be open sourced as part of the Delta Lake 2.0 release. In addition, the company has announced a new version of its MLflow machine learning platform, which includes MLflow Pipeline, a new feature that simplifies the development of machine learning models.



Databricks was founded in 2013. The company invented the concept of a 'data lakehouse', which combines structured data storage in a data warehouse with unstructured data storage in a data lake. This combined method simplifies the use of data for the deployment of machine learning models.

The concept has proven popular, and Databricks claims to be used by over 7,000 organisations worldwide, including 40 per cent of the Fortune 500. It makes money by selling subscriptions to analytics tools that are used on top of Delta Lake.

Darktable 4.0 also includes support for Filmic v6. True gamut mapping replaces mandatory desaturation near medium white and black in this new colour science. According to Obry, the result is more saturated colours and bluer skies. Filmic v6 also includes what Darktable refers to as a 'hue handcuff', which prevents hue shifts that can degrade. It claims that this helps keep skies blue rather than shifting to cyan, and keeps reds from fading into a warmer yellow.

Darktable 4.0 also includes the 'guided laplacian' method of highlight reconstruction. Users can use an iterative and multi-scale wavelet scheme to extract more information from RGB channels. When the colour is clipped, this laplacian technique prevents colour bleeding. This feature is not available for X-Trans sensor images, only Bayer sensor images. Uniform Color Space 2022 is also included in the update (USC22).

Darktable's user interface has also been completely rewritten in version 4.0. According to Obry, the app has been redesigned to improve its look and consistency, and padding, margins, colour, contrast, alignment, and icons have all been reworked. Obry also mentions that collapsible sections within modules have been redesigned to be more obvious and visually appealing, and that multiple sections and modules have been cleaned up and adjusted for clarity. Darktable 4.0 also includes improvements to performance and OpenCL support.

Ant Group's Computation Framework is now open source

Ant Group has announced that its privacy-preserving Computation Framework (or 'Framework') has gone open source, with the goal of making the technologies more accessible to global developers and speeding up the Framework's application in various scenarios. The Framework is intended to address issues that arise during multi-party collaboration on information flows. It incorporates a number of industry-leading privacy computing technologies from Ant Group and covers the entire information life cycle, including secure multi-party computation, differential privacy, homomorphic encryption, and others.

The 'CCF-Ant Group Privacy Computing Special Fund' (or the 'Fund') has also been officially launched by Ant Group and the China Computer Federation. The Fund will initially invest over US\$ 600,000 in cutting-edge research in privacy computing fields such as post-quantum multi-party computation, security of open source privacy computing platform, and others. The Fund is now available to researchers all over the world.

